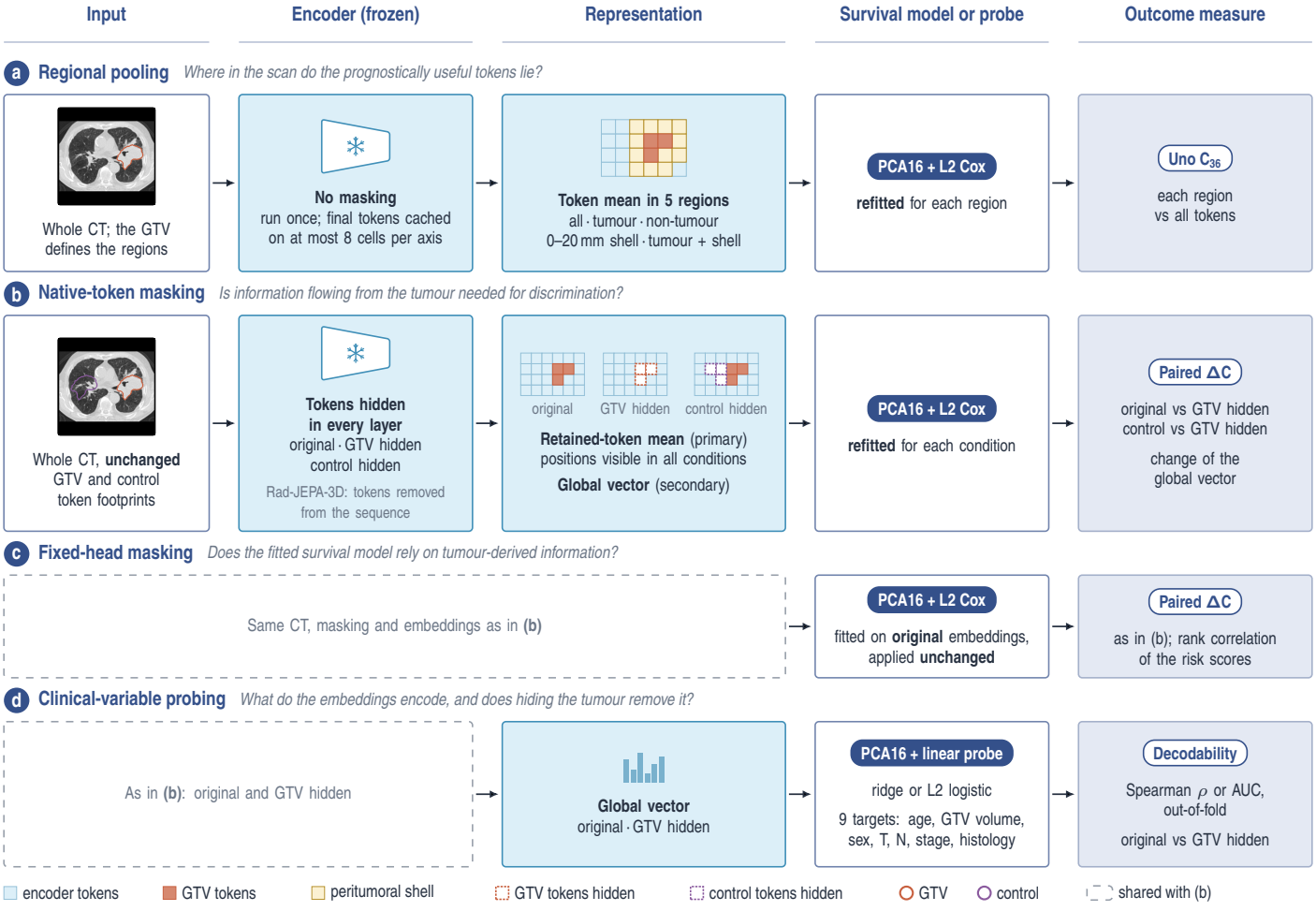

Input
Encoder (frozen)
Representation
Survival model or probe
Outcome measure
a Regional pooling
Where in the scan do the prognostically useful tokens lie?
Whole CT; the GTV defines the regions
No masking
run once; final tokens cached on at most 8 cells per axis
Token mean in 5 regions
all · tumour · non-tumour
0–20 mm shell · tumour + shell
PCA16 + L2 Cox
refitted for each region
Uno C36
each region vs all tokens
b Native-token masking
Is information flowing from the tumour needed for discrimination?
Whole CT, unchanged GTV and control token footprints
Tokens hidden in every layer
original · GTV hidden
control hidden
Rad-JEPA-3D: tokens removed from the sequence
original
GTV hidden
control hidden
Retained-token mean (primary)
positions visible in all conditions
Global vector (secondary)
PCA16 + L2 Cox
refitted for each condition
Paired ΔC
original vs GTV hidden
control vs GTV hidden
change of the global vector
c Fixed-head masking
Does the fitted survival model rely on tumour-derived information?
Same CT, masking and embeddings as in (b)
PCA16 + L2 Cox
fitted on original embeddings, applied unchanged
Paired ΔC
as in (b); rank correlation of the risk scores
d Clinical-variable probing
What do the embeddings encode, and does hiding the tumour remove it?
As in (b): original and GTV hidden
Global vector
original · GTV hidden
PCA16 + linear probe
ridge or L2 logistic
9 targets: age, GTV volume, sex, T, N, stage, histology
Decodability
Spearman ρ or AUC, out-of-fold
original vs GTV hidden
encoder tokens
GTV tokens
peritumoral shell
GTV tokens hidden
control tokens hidden
GTV
control
shared with (b)